

Database Layout

Field	Value
Purpose & Intent	Document the persistent data model for players, maps, blueprints, component catalog, mount points, installed blueprint components, vehicles, and map placement.
Incoming	DTO structs in <code>src/backend/src/</code> and SQL migrations in <code>src/backend/migrations/</code>
Outgoing	Ch. 5 Building Block View (backend component), Ch. 6 Runtime View (data-access scenarios), Ch. 9 Architecture Decisions (ADR-001)

Scope

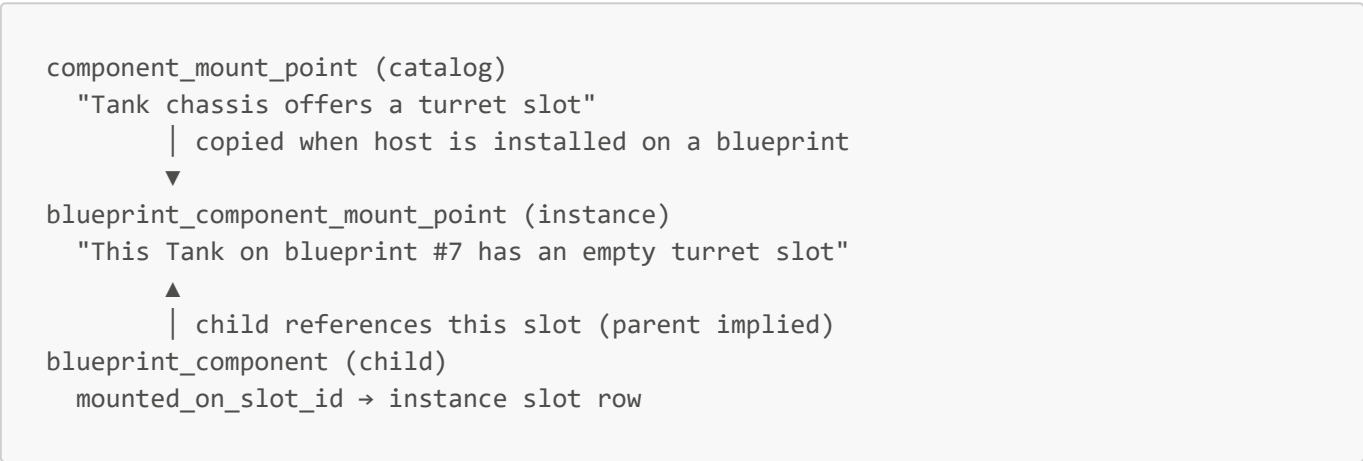
This document describes the **target schema** for blueprint assembly (catalog + instance mount points). See [Migration drift](#) for differences from the current SQL migration.

Three layers:

Layer	Tables	Role
Catalog	<code>component_definition</code> , <code>component_mount_point</code>	Global, seeded part types and the slots each host type offers
Blueprint design	<code>blueprint</code> , <code>blueprint_component</code> , <code>blueprint_component_mount_point</code>	Player-owned design: installed parts and instance slots copied from catalog
Fleet / map	<code>vehicle</code> , <code>vehicle_on_map</code>	Bought units and map placement

Mount rule (v1): a child may attach when `child.kind = blueprint_component_mount_point.accepts_kind` on an **empty** instance slot of the parent host. See [TW-1](#).

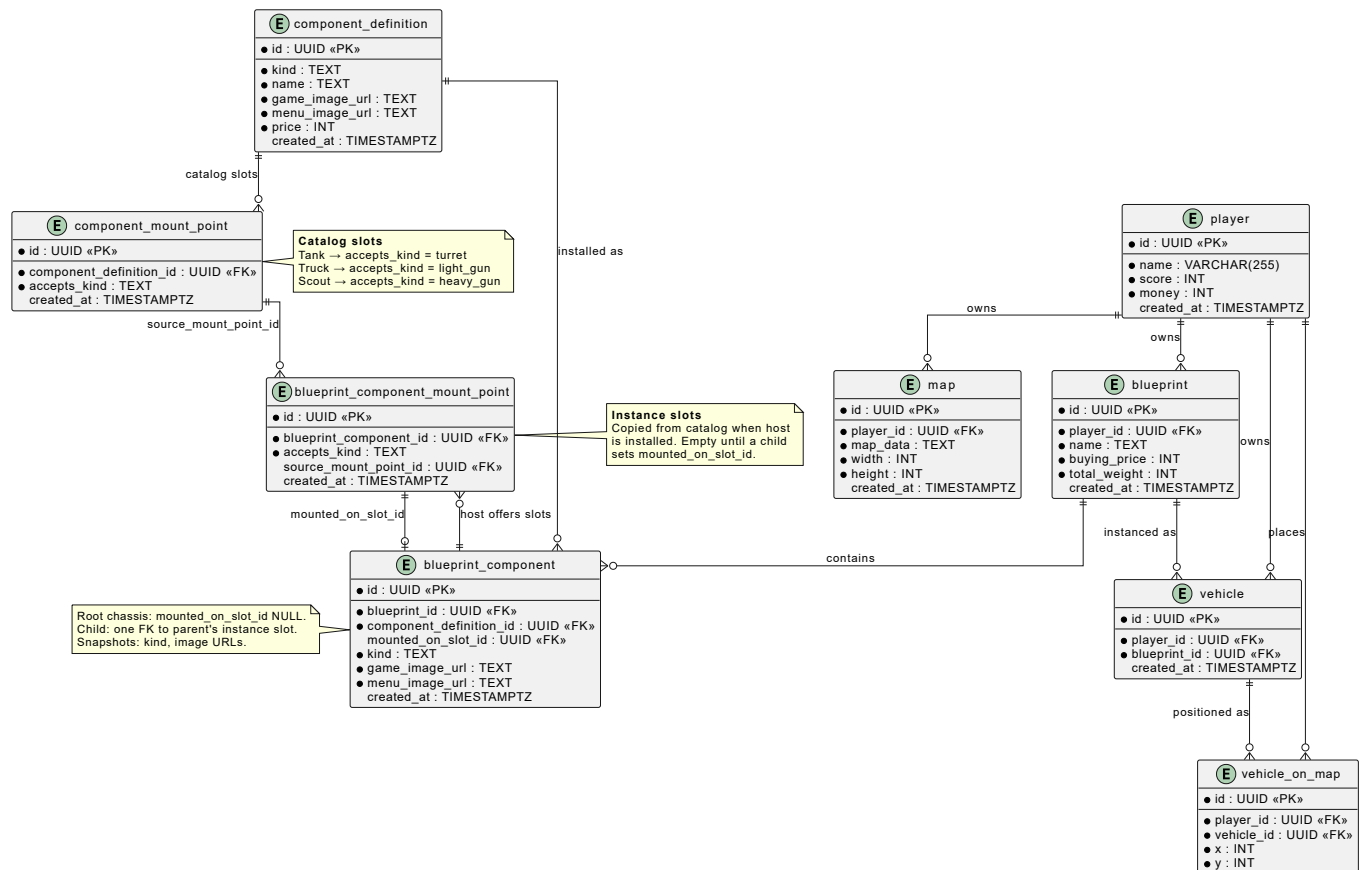
Catalog vs. instance mount points



- **Catalog** (`component_mount_point`) — shared by all players; defines what a part type *can* offer.

- **Instance** (`blueprint_component_mount_point`) — per installed host on one blueprint; defines what *this* chassis/turret offers right now.
- **Child** (`blueprint_component`) — points at the parent's **instance slot**, not at the catalog. The parent does not reference children.

Entity-Relationship Diagram



Tables

player

Column	Type	Constraints	Notes
id	UUID	PK, NOT NULL	
name	VARCHAR(255)	NOT NULL	
score	INT	NOT NULL	
money	INT	NOT NULL	
created_at	TIMESTAMPZ	DEFAULT NOW()	

map

Column	Type	Constraints	Notes
id	UUID	PK, NOT NULL	

Column	Type	Constraints	Notes
player_id	UUID	FK → player.id, NOT NULL	Owning player
map_data	TEXT	NOT NULL	Serialised map content
width	INT	NOT NULL	
height	INT	NOT NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

blueprint

Column	Type	Constraints	Notes
id	UUID	PK, NOT NULL	
player_id	UUID	FK → player.id, NOT NULL	Owning player
name	TEXT	NOT NULL	Blueprint name
buying_price	INT	NOT NULL	Cached total cost
total_weight	INT	NOT NULL	Cached total weight
created_at	TIMESTAMPTZ	DEFAULT NOW()	

component_definition

Global **parts catalog**: one row per buyable/mountable part type.

Seeded kinds (v1): chassis, turret, light_gun, heavy_gun.

Column	Type	Constraints	Notes
id	UUID	PK, NOT NULL	
kind	TEXT	NOT NULL	Part category; used for mount matching
name	TEXT	NOT NULL	Human-readable name
game_image_url	TEXT	NOT NULL	Frontend asset path for gameplay
menu_image_url	TEXT	NOT NULL	Frontend asset path for menus
price	INT	NOT NULL	Purchase price
created_at	TIMESTAMPTZ	DEFAULT NOW()	

component_mount_point

Catalog slots — what a host part type offers globally.

Column	Type	Constraints	Notes
id	UUID	PK, NOT NULL	

Column	Type	Constraints	Notes
<code>component_definition_id</code>	UUID	FK → <code>component_definition.id</code> , NOT NULL	Catalog host (e.g. Tank chassis)
<code>accepts_kind</code>	TEXT	NOT NULL	Required <code>kind</code> on the child part
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Seeded catalog graph (v1):

Host	<code>accepts_kind</code>
Tank (<code>chassis</code>)	<code>turret</code>
Truck (<code>chassis</code>)	<code>light_gun</code>
Scout (<code>turret</code>)	<code>heavy_gun</code>

`blueprint_component`

Installed part on a player blueprint.

Column	Type	Constraints	Notes
<code>id</code>	UUID	PK, NOT NULL	
<code>blueprint_id</code>	UUID	FK → <code>blueprint.id</code> , NOT NULL	Owning blueprint
<code>component_definition_id</code>	UUID	FK → <code>component_definition.id</code> , NOT NULL	Installed part from catalog
<code>mounted_on_slot_id</code>	UUID	FK → <code>blueprint_component_mount_point.id</code> , NULL	Parent's instance slot; NULL for root chassis
<code>kind</code>	TEXT	NOT NULL	Snapshot from catalog at install
<code>game_image_url</code>	TEXT	NOT NULL	Snapshot
<code>menu_image_url</code>	TEXT	NOT NULL	Snapshot
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

When a host part is installed, copy its catalog mount points into `blueprint_component_mount_point` rows owned by that `blueprint_component`.

`blueprint_component_mount_point`

Instance slot on an installed host — copied from catalog at install time.

Column	Type	Constraints	Notes
<code>id</code>	UUID	PK, NOT NULL	
<code>blueprint_component_id</code>	UUID	FK → blueprint_component.id, NOT NULL	Installed host that offers this slot
<code>accepts_kind</code>	TEXT	NOT NULL	Copied from catalog; child must match
<code>source_mount_point_id</code>	UUID	FK → component_mount_point.id, NULL	Optional traceability to catalog row
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Slot occupancy: a slot is empty when no `blueprint_component` row has `mounted_on_slot_id` pointing at it. At most one child per instance slot.

Finding the parent host from a child: `child.mounted_on_slot_id` → `blueprint_component_mount_point.blueprint_component_id`.

vehicle

Column	Type	Constraints	Notes
<code>id</code>	UUID	PK, NOT NULL	
<code>player_id</code>	UUID	FK → player.id, NOT NULL	Owning player
<code>blueprint_id</code>	UUID	FK → blueprint.id, NOT NULL	Blueprint this vehicle was built from
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

vehicle_on_map

Column	Type	Constraints	Notes
<code>id</code>	UUID	PK, NOT NULL	
<code>player_id</code>	UUID	FK → player.id, NOT NULL	Map owner
<code>vehicle_id</code>	UUID	FK → vehicle.id, NOT NULL	Placed vehicle
<code>x</code>	INT	NOT NULL	Grid x
<code>y</code>	INT	NOT NULL	Grid y

Constraints & indexes (target)

Apply in the follow-up migration that introduces instance mount points:

Table	Constraint	Purpose
<code>component_mount_point</code>	UNIQUE (<code>component_definition_id</code> , <code>accepts_kind</code>)	One catalog slot per kind per host

Table	Constraint	Purpose
<code>blueprint_component_mount_point</code>	UNIQUE (<code>blueprint_component_id</code> , <code>accepts_kind</code>)	One instance slot per kind per installed host
<code>blueprint_component</code>	Index on <code>mounted_on_slot_id</code>	Child lookup and slot occupancy checks

Slot occupancy is enforced in application logic (no child may reference the same `blueprint_component_mount_point.id`); a partial unique index on `mounted_on_slot_id` WHERE NOT NULL is optional if the DB should enforce it.

Install flow (v1)

1. Buy chassis (root)

- Insert `blueprint_component` with `mounted_on_slot_id = NULL`.
- For each `component_mount_point` on the chassis definition, insert `blueprint_component_mount_point` on that row.

2. Mount child (e.g. turret on Tank)

- Find an instance slot on the parent host where `accepts_kind = child.kind` and slot is empty.
- Insert `blueprint_component` with `mounted_on_slot_id` → that slot.
- Copy catalog mount points from the child's definition → new instance slots on the child row.

3. Validation

- `child.kind` must equal target slot's `accepts_kind`.
- Target slot must have no existing child (`mounted_on_slot_id` unused).

Assembly examples

Tank blueprint

```
bc1 blueprint_component (Tank chassis)      mounted_on_slot_id: NULL
  bcmp1 instance slot (accepts turret)      ← empty, then occupied
    bc2 blueprint_component (Scout)         mounted_on_slot_id → bcmp1
      bcmp2 instance slot (accepts heavy_gun)
        bc3 blueprint_component (Main Gun) mounted_on_slot_id → bcmp2
```

Truck blueprint

```
bc1 blueprint_component (Truck chassis)      mounted_on_slot_id: NULL
  bcmp1 instance slot (accepts light_gun)
    bc2 blueprint_component (Light MG)      mounted_on_slot_id → bcmp1
```

DTO Mapping

Table	DTO struct	Notable differences
player	PlayerDto	money and score exposed
map	MapDto	created_at as Option<String>
blueprint	BlueprintDto	player_id, name, buying_price, total_weight
component_definition	ComponentDefinitionDto	Image URLs and price
component_mount_point	none yet	Catalog; read when copying instance slots
blueprint_component	none yet	Assembly; chassis resolved via kind = 'chassis' in vehicle DTOs today
blueprint_component_mount_point	none yet	Instance slots; drives empty-slot UI
vehicle	VehicleDto (partial)	Chassis image only until TW-1 assembly DTO
vehicle_on_map	VehicleOnMapDto	Position + display fields

Migration drift

The migration `20260529195524_create_tank_wars_table.up.sql` **does not yet match** this target. Current code has:

Current	Target
<code>blueprint_component.mount_point_id</code>	Remove — use instance layer
<code>blueprint_component.parent_component_mount_point_id</code>	Remove — use <code>mounted_on_slot_id</code>
(missing)	Add <code>blueprint_component_mount_point</code>
(missing)	Add <code>blueprint_component.mounted_on_slot_id</code>

A follow-up migration should introduce `blueprint_component_mount_point` and replace the dual catalog FKs on `blueprint_component`. Apply the [constraints & indexes](#) when doing so.

Related docs: [ADR-001](#) (decision rationale), [TW-1](#) (implementation plan).

Notes

- Catalog mount points are seeded in `src/backend/src/seed.rs`; instance slots are created at install time.
- Denormalized fields on `blueprint_component` are install-time snapshots.
- Empty-slot UI reads `blueprint_component_mount_point` for a selected host and excludes slots already referenced by `mounted_on_slot_id`.
- DAO layer owns reads/writes; DTOs remain API shapes only.